Alternative QFT

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The hidden subgroup problem (HSP) yielding exponential quantum speedup over the best known classical algorithms, uses the Quantum Fourier Transform (QFT). Standard HSP algorithms require the full quantum Fourier transform (QFT), whose circuit depth and long-range gates remain challenging for near-term hardware. We propose a departure from conventional QFT approximations: instead of mimicking its gate structure, we identify two functional properties that are exploited by the QFT to enable HSP. Firstly the shift invariance property of the QFT allows for the removal of a random overall shift. Secondly, the information about the hidden subgroup generator needs to remain accessible in the computational basis, and we quantify that via the Fisher information. We construct a family of shallow circuits that preserve shift invariance. Numerical analysis shows these circuits retain Fisher information. We demonstrate successful reconstruction using a neural network and analyze performance under noise, outlining a potential path to practical HSP implementations.

Introduction. — The hidden subgroup problem (HSP) serves as a central source of exponential quantum speedups, establishing Fourier-type interference as a foundational primitive in quantum algorithms. Traditionally, standard HSP solvers rely heavily on the quantum Fourier transform (QFT) to extract subgroup information. However, implementing the full QFT necessitates long-range controlled-phase operations and entails a nontrivial circuit depth, both of which present significant challenges for near-term experimental realizations.

To mitigate these hardware demands, most existing approaches aim to reduce the operational cost by constructing approximate-QFT circuits. Nevertheless, this line of work predominantly treats the QFT itself as the ultimate target, limiting its scope primarily to gate-level simplifications. In this work, we adopt a fundamentally different perspective: rather than attempting to approximate the QFT gate by gate, we investigate which intrinsic structural properties of the transformation are genuinely responsible for extracting hidden subgroup information.

Specifically, we focus on two critical properties: shift invariance and Fisher information. Shift invariance characterizes whether the interference pattern induced by subgroup cosets is effectively preserved in the measurement statistics, and in particular whether the random coset representative can be removed at the probability level. This alone is not sufficient: after measurement, one must still be able to recover the hidden subgroup generator from the observed samples. In the regimes considered here, we treat that generator as the parameter encoded in the output distribution, and use the Fisher information to quantify how much information the samples retain about it. Driven by these structural requirements, we identify a distinct family of quantum circuits that strictly satisfies the shift-invariance condition while retaining sufficiently high Fisher information to enable reliable reconstruction of the hidden subgroup using a neural network. Consequently, this circuit family serves as a principled alternative to the QFT, moving beyond purely heuristic approximations.

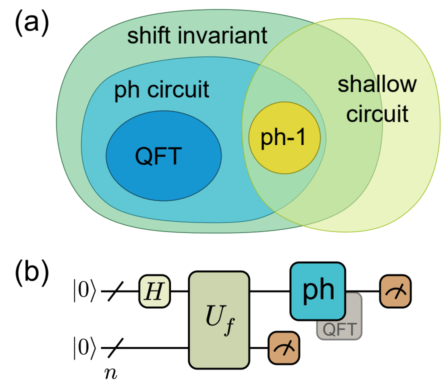

FIG. 1. **Conceptual framework of the shift-invariant HPcircuit.** (a) Relationship between different families of quantum transformations. The proposed HPcircuit belongs to the broader class of shift-invariant circuits and serves as a shallow-depth alternative to the standard quantum Fourier transform (QFT). (b) Schematic of the hidden subgroup problem (HSP) solver. Following the oracle U_f evaluation, the traditional QFT block is replaced by our shift-invariant HPcircuit to extract the subgroup information before measurement.

To evaluate the viability of this approach, we analytically compute the Fisher information of the standard QFT as a benchmark, with the derivation given in Appendix D. We then numerically investigate how the Fisher information of our proposed circuit family scales with both the number of qubits and the circuit depth. This analysis provides critical evidence regarding whether the hidden subgroup can still be recovered using feasible classical post-processing. Our experiments indicate that the Fisher information remains substantial across the tested parameter regimes, suggesting that even relatively shallow circuits can retain sufficient information for hidden subgroup recovery.

We employ a neural network as a classical post-processing module, demonstrating that the hidden subgroup can indeed be successfully reconstructed directly from the corresponding measurement data. Finally, we

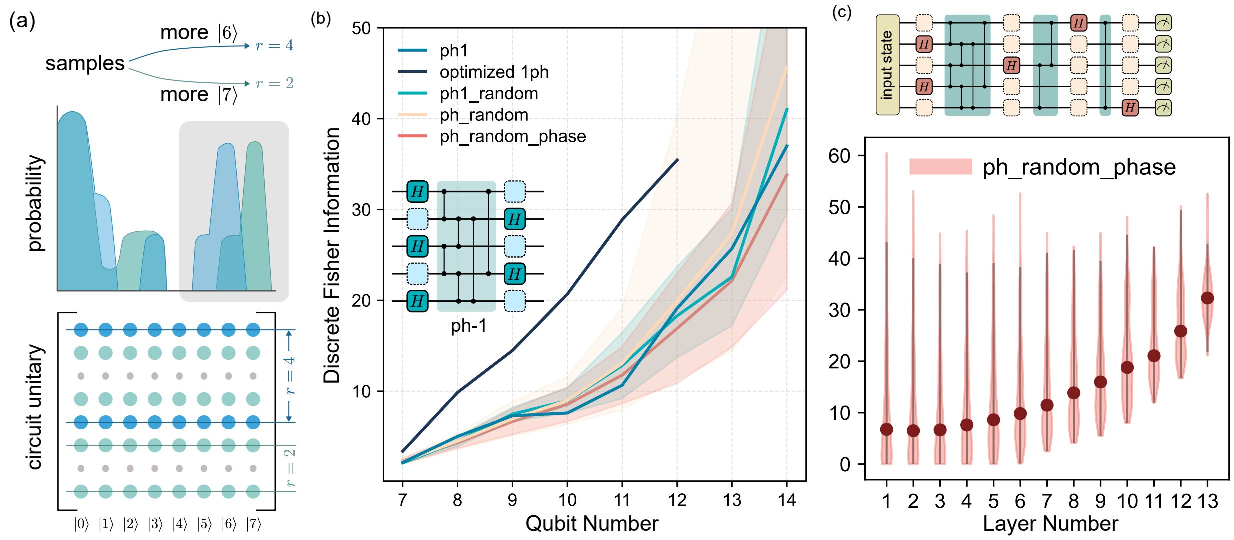

FIG. 2. Fisher Information scaling for candidate circuits.

analyze the robustness of the proposed circuits under a depolarizing noise model, with detailed settings and full numerical results deferred to the Appendix.

Necessary condition for shift invariance. — To establish a precise foundation for the role of Fourier-type transformations, we first review the mathematical definition of the hidden subgroup problem (HSP) alongside the standard quantum routine employed for its resolution. Within the context of this routine, we elucidate how shift invariance serves as the fundamental physical property enabling the controlled, information-preserving interference of distinct hidden subgroup cosets.

Definition 1 (Shift invariance)**.** *A quantum circuit U is shift-invariant with respect to an abelian group G if for any input state $|\psi\rangle$ and any group element $g \in G$, the output state satisfies $U|\psi\rangle = U|\psi + g\rangle$, where the addition is defined according to the group operation.*

Lemma 1. *A unitary operator U is shift-invariant only if the matrix elements of U satisfy the condition $U_{ij} = 1/\sqrt{|G|}e^{\mathrm{i}\theta_{ij}}$, where θ and group size $|G|$.*

the proof of this lemma is provided in the Appendix.

Building upon this conceptual framework, we then introduce a formal definition of a shift-invariant circuit, detailing how this mathematical formulation encapsulates the critical operational requirements of the HSP measurement stage. Based on this definition, we derive the strict necessary conditions that any prospective candidate circuit must fulfill. This derivation systematically narrows the search space for viable alternatives to the quantum Fourier transform (QFT), thereby illuminating the underlying structural properties requisite for successful circuit constructions.

HP: A shift invariant family of circuits. — To bridge the gap between the necessary conditions for shift invariance and explicit circuit construction, we introduce a parameterized family of quantum circuits, denoted as the HP-k circuits. The fundamental design principle of this architecture is to systematically interleave transversal layers of single-qubit Hadamard gates with parameterized controlled-phase operations. By doing so, the circuit hierarchically builds the precise phase relationships required for subgroup interference.

Let H_i denote the Hadamard gate applied to qubit i. We partition the circuit into L sequential stages. For a given stage l, we define a subset of target qubits Λ_l and a counterpart subset of control qubits $\bar{\Lambda}_l$. These subsets are coupled via a controlled-phase block, denoted as $\mathrm{CP}_{\Lambda_l,\bar{\Lambda}_l}(\boldsymbol{\theta}_{\Lambda_l\bar{\Lambda}_l})$, which applies pairwise operations $\mathrm{CP}_{jk}(\theta_{jk})$ between each qubit $j \in \bar{\Lambda}_l$ and $k \in \Lambda_l$. Building upon these notations, we formally define the circuit architecture:

Definition 2 (HP-k circuit)**.** *A quantum circuit is classified as an HP-k circuit if it is constructed using the following architecture:*

$$U_{\mathrm{HP}}^{(L)}(\boldsymbol{\theta}) = \bigotimes_{i\in\Lambda_L} H_i \prod_{l=L-1}^{1} \left(\mathrm{CP}_{\Lambda_l,\bar{\Lambda}_l}(\boldsymbol{\theta}_{\Lambda_l\bar{\Lambda}_l}) \bigotimes_{i\in\Lambda_l} H_i \right), \tag{1}$$

where the controlled-phase block is given by:

$$\mathrm{CP}_{\Lambda_l,\bar{\Lambda}_l}(\boldsymbol{\theta}_{\Lambda_l\bar{\Lambda}_l}) = \prod_{j\in\bar{\Lambda}_l} \prod_{k\in\Lambda_l} \mathrm{CP}_{jk}(\theta_{jk}). \tag{2}$$

By instantiating this architecture, we identify a concrete circuit family that strictly satisfies the previously derived shift-invariance conditions, thereby providing an explicit and nontrivial replacement for the standard QFT circuit. Importantly, this proposed family encompasses a strictly broader set of transformations than those generated by conventional QFT constructions. This generalization demonstrates that shift invariance is not a unique feature of the QFT, but rather a robust property that can naturally arise within a significantly larger design space of practically useful quantum circuits.

Theorem 2. *HP-k circuits are shift invariant under group* $\prod_i \mathbb{Z}_{2^{m_i}}$*.*

HP-0 for hidden subgroup on $\mathbb{Z}_{2^n}$*.*— We first consider the cyclic group $\mathbb{Z}_{2^n}$, which is naturally represented by an n-qubit computational basis and therefore provides a clean setting for isolating the effect of the circuit transformation. In this case, the subgroup structure is particularly simple: each subgroup is generated by a power of two, and the corresponding coset states have a regular binary structure. For example, if $H = \{q2^p : q = 0, \ldots, 2^{n-p} - 1\}$, then, up to normalization and bit-ordering convention, applying a transversal Hadamard layer to a coset state gives

$$H^{\otimes n} \sum_{h \in H} |c + h\rangle \propto \sum_{k \in \{0,1\}^p} (-1)^{k \cdot c} |0\rangle^{\otimes(n-p)} |k\rangle . \quad (3)$$

Thus the dependence on the coset representative c appears only through phases, while the support of the measurement distribution reveals the number of fixed zero bits and hence the hidden subgroup parameter. The detailed calculation, together with the corresponding sampling algorithm using transversal Hadamard gates, is provided in Appendix C. A transversal layer of Hadamard gates, which we identify as the HP-0 circuit, is therefore already sufficient to remove the irrelevant coset shift at the level of measurement probabilities and to retain information about the hidden subgroup in the idealized finite-group setting. Thus, HP-0 provides a useful baseline demonstrating that a shallow shift-invariant transformation can be sufficient for a meaningful subclass of HSP instances.

At the same time, the $\mathbb{Z}_{2^n}$ case should be regarded as an instructive but restricted regime. Its subgroup structure is highly constrained, and several such instances admit efficient classical treatment, so this setting alone does not capture the full algorithmic role played by the QFT. The more demanding regime arises when one uses a finite-dimensional register to approximate a problem over $\mathbb{Z}$ and to recover a subgroup such as $\mathbb{Z}_q$. In that situation, the sampled state need not be an exact subgroup state of the truncated register, and the robustness of the standard QFT becomes essential. This motivates the extension from HP-0 to deeper members of the HPfamily, which are designed to preserve shift-invariant sampling structure while allowing greater expressive power than a single Hadamard layer.

Retention of Fisher information on hidden subgroup.—

To justify the post-processing approach more systematically, we ask whether the hidden subgroup generator can be estimated using only a polynomial number of measurement samples. In our setting, this generator plays the role of the parameter in the output distribution, so the relevant question is how sensitive the measured probabilities are to changes in that generator. Fisher information provides a natural metric for this purpose, since it quantifies the amount of recoverable information about the generator contained in the samples. As a benchmark, for the standard QFT we obtain

$$I_{\mathrm{QFT}}(r) = \frac{8}{\pi R} \sum_{k=1}^{R-1} (R-k) \left[\frac{k\pi^3}{3} + \sum_{j=1}^{k-1} \frac{k-j}{j^3 r^3} \Gamma_j(r) \right],$$
$$\Gamma_j(r) = 2\pi jr \cos(2\pi jr) + (2\pi^2 j^2 r^2 - 1) \sin(2\pi jr), \quad (4)$$

which in the large-r regime reduces to $I_{\mathrm{QFT}}(r) \approx \frac{4\pi^2}{9}(R^2 - 1)$. The detailed derivation is given in Appendix D.

HP-1 retains Fisher information with fewer layers than QFT.. — We compute the Fisher information numerically as a function of both circuit depth and system size. The observed scaling offers strong evidence that the proposed circuit family preserves enough information for practical inference. Most notably, HP-1 retains a level of Fisher information comparable to that of the full QFT while remaining considerably shallower, thereby reducing the depth requirements of the transformation stage.

Simulation of HSP with HP-1 and efficient neural net post-processing. — We use a neural network as a learnable classical post-processing layer that takes measurement outcomes from the HP-1 circuit and predicts the hidden subgroup parameter. This experiment is intended to test the practical recoverability of subgroup information, rather than only reporting information-theoretic quantities in the abstract. The resulting performance shows that the measurement distributions produced by these shallow circuits still contain highly usable structure, and that an efficient classical decoder can successfully exploit it.

Robustness under noise.— We also evaluate the previously trained HP-1 decoder under a post-circuit global depolarizing channel. The held-out recovery accuracy remains essentially unchanged up to noise strength $\eta \approx 0.215$, while stronger noise causes a marked drop: at $\eta = 0.464$ the accuracy is 0.491 for 9 qubits and 0.881 for 10 qubits, and at the fully depolarizing point $\eta = 1$ it falls to chance. Detailed settings, complete numerical results, and the corresponding interpretation are given in Appendix E.

Summary and outlook. — In summary, we have established a framework for solving the Hidden Subgroup Problem using shift-invariant circuits that circumvent the depth and gate-connectivity bottlenecks of the traditional QFT. The main remaining limitation of the current proposal is its generalization ability. Extending this approach, together with the corresponding classical post-processing, to more complex group structures requires further investigation and will form the basis of our future work.

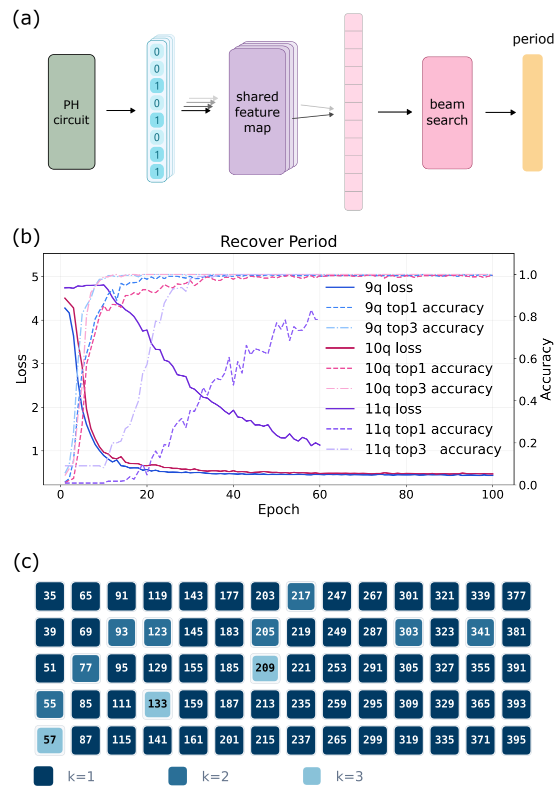

FIG. 3. Subgroup recovery using a neural network over measurement outcomes.

APPENDIX

A. Hidden Subgroup Problem and Fourier Sampling

Many significant quantum algorithms, including Shor's algorithm, are special instances of the HSP.

Definition 3 (Hidden Subgroup Problem)**.** *Let G be a group and $H \leq G$ be an unknown subgroup. We are given oracle access to a function $f : G \to X$. The function f has the property that for any $g_1, g_2 \in G$:*

$$f(g_1) = f(g_2) \iff g_1 g_2^{-1} \in H \tag{A1}$$

In other words, f is constant on the right cosets of H and distinct on different cosets. The goal is to find a hidden subgroup H.

For the HSP of finite abelian group G, like Shor algorithm, there is a standard algorithm that solves the problem efficiently. For simplicity and convenience, we will assume that the group G we discussed in this section is finite and abelian.

Group theory reveals that all finite abelian groups are isomorphic to a direct product of cyclic groups,

$$G = \mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \cdots \times \mathbb{Z}_{n_k} \tag{A2}$$

B. Necessary condition of shift invariance

Here we make explicit the probability-level shift-invariance condition used in Fourier sampling. Let G be a finite abelian group with $N = |G|$, and let $\{|g\rangle : g \in G\}$ denote the computational basis. For $c \in G$, define the shift operator

$$T_c |g\rangle = |g + c\rangle . \tag{B1}$$

For a subgroup $H \leq G$, write the normalized subgroup state as

$$|H\rangle = \frac{1}{\sqrt{|H|}} \sum_{h \in H} |h\rangle\,. \tag{B2}$$

The shift-invariance requirement relevant to HSP sampling is that the measurement distribution after applying U is independent of the coset representative:

$$|\langle x| U |H\rangle|^2 = |\langle x| UT_c |H\rangle|^2\,, \qquad \forall x, c \in G,\ \forall H \leq G. \tag{B3}$$

Thus $U|H\rangle$ and $U|c+H\rangle$ may differ by phases, but they must produce the same probabilities in the computational basis. The necessary modulus condition in Lemma 1 follows by applying Eq. (B3) to the trivial subgroup.

Proof of Lemma 1. Take the trivial subgroup $H = \{0\}$. Then $|H\rangle = |0\rangle$ and $T_c|H\rangle = |c\rangle$ for every $c \in G$. Equation (B3) therefore implies

$$|\langle x| U |0\rangle|^2 = |\langle x| U |c\rangle|^2\,, \qquad \forall x, c \in G. \tag{B4}$$

With the convention $U_{xc} = \langle x| U |c\rangle$, this says that for each fixed output label x, all entries in the x-th row of U have the same squared modulus. Denote this common value by a_x. Since U is unitary, each row is normalized:

$$1 = \sum_{c \in G} |U_{xc}|^2 = N a_x. \tag{B5}$$

Hence $a_x = 1/N$ for every $x \in G$. Therefore every matrix element satisfies

$$|U_{xc}| = \frac{1}{\sqrt{N}} = \frac{1}{\sqrt{|G|}}. \tag{B6}$$

Writing each nonzero complex number in polar form gives

$$U_{xc} = \frac{1}{\sqrt{|G|}} e^{\mathrm{i}\theta_{xc}}, \qquad \theta_{xc} \in \mathbb{R}. \tag{B7}$$

This proves the claimed necessary condition. □

C. Transversal Hadamard sampling and efficiency guarantee

We now give the calculation underlying the HP-0 discussion in the main text. Consider $G = \mathbb{Z}_{2^n}$ represented by n computational-basis qubits, and use the bit ordering in which a multiple of 2^p has the form

$$h = h_1 h_2 \cdots h_{n-p} 0 \cdots 0. \tag{C1}$$

The subgroup generated by 2^p is

$$H_p = \{q 2^p : q = 0, \ldots, 2^{n-p} - 1\}, \tag{C2}$$

so $|H_p| = 2^{n-p}$. Since adding an element of H_p can vary the first $n-p$ bits, each coset has a representative of the form

$$c = 0 \cdots 0 c_{n-p+1} \cdots c_n. \tag{C3}$$

Thus a normalized coset state can be written as

$$|c + H_p\rangle = \frac{1}{\sqrt{2^{n-p}}} \sum_{h_1, \ldots, h_{n-p} \in \{0,1\}} |h_1 \cdots h_{n-p} c_{n-p+1} \cdots c_n\rangle\,. \tag{C4}$$

For an n-bit string x, the transversal Hadamard transform satisfies

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{k \in \{0,1\}^n} (-1)^{k \cdot x} |k\rangle\,. \tag{C5}$$

Applying this to Eq. (C4) gives

$$\begin{aligned} H^{\otimes n}\left|c+H_p\right\rangle &= \frac{1}{\sqrt{2^{n-p}}}\frac{1}{\sqrt{2^n}}\sum_{k\in\{0,1\}^n}\sum_{h_1,\ldots,h_{n-p}\in\{0,1\}}(-1)^{\sum_{r=1}^{n-p}k_rh_r+\sum_{r=n-p+1}^{n}k_rc_r}\left|k\right\rangle \\ &= \frac{1}{\sqrt{2^{n-p}}}\frac{1}{\sqrt{2^n}}\sum_{k\in\{0,1\}^n}(-1)^{\sum_{r=n-p+1}^{n}k_rc_r}\prod_{r=1}^{n-p}\left(\sum_{h_r\in\{0,1\}}(-1)^{k_rh_r}\right)\left|k\right\rangle . \end{aligned} \tag{C6}$$

The inner product factor is zero unless $k_1=\cdots=k_{n-p}=0$, in which case it equals 2^{n-p}. Therefore

$$H^{\otimes n}\left|c+H_p\right\rangle = \frac{1}{\sqrt{2^p}}\sum_{k_{n-p+1},\ldots,k_n\in\{0,1\}}(-1)^{\sum_{r=n-p+1}^{n}k_rc_r}\left|0\right\rangle^{\otimes(n-p)}\left|k_{n-p+1}\cdots k_n\right\rangle . \tag{C7}$$

The coset representative c only contributes the phase factor in Eq. (C7). Consequently, the measurement distribution is independent of c: the first $n-p$ bits are always zero, while the remaining p bits are uniformly random.

This immediately gives a simple reconstruction procedure for the idealized $\mathbb{Z}_{2^n}$ HSP. Prepare independent coset states using the HSP oracle, apply $H^{\otimes n}$ to the first register, and measure in the computational basis. For each bit position, record whether a 1 ever appears. A bit that is always zero is classified as fixed by the subgroup, while a bit for which a 1 appears at least once is classified as free. The number of free trailing bits determines p, and hence determines $H_p=\langle 2^p\rangle$ under the bit-ordering convention above.

The sample complexity is logarithmic in the number of qubits. Fixed bits are never misclassified, because they are exactly zero in every sample. For a free bit, the probability that it appears as zero in all m independent samples is 2^{-m}. By the union bound, the probability of misclassifying at least one free bit is at most

$$p2^{-m}\le n2^{-m}. \tag{C8}$$

Thus $m\ge\lceil\log_2(n/\epsilon)\rceil$ samples are sufficient to identify the fixed and free bit positions, and therefore the hidden subgroup H_p, with probability at least $1-\epsilon$.

D. QFT Fisher information for subgroup-generator estimation

To benchmark the proposed circuit family, we treat the hidden subgroup generator r as the parameter to be inferred from the measurement distribution produced by the standard QFT. Consider the truncated periodic state

$$\left|\psi_r\right\rangle = \frac{1}{\sqrt{R}}\sum_{q=0}^{R-1}\left|qr\right\rangle , \tag{D1}$$

where R is the number of multiples of r supported in the 2^n-dimensional register. For the 2^n-point QFT,

$$\left|j\right\rangle \mapsto \frac{1}{\sqrt{2^n}}\sum_{k=0}^{2^n-1}e^{2\pi\mathrm{i}jk/2^n}\left|k\right\rangle , \tag{D2}$$

and therefore

$$\left|\psi_r\right\rangle \mapsto \frac{1}{\sqrt{R2^n}}\sum_{k=0}^{2^n-1}\left(\sum_{q=0}^{R-1}e^{2\pi\mathrm{i}qrk/2^n}\right)\left|k\right\rangle . \tag{D3}$$

The corresponding measurement probability is

$$p(k|r) = \frac{1}{R2^n}\left|\sum_{q=0}^{R-1}e^{2\pi\mathrm{i}qrk/2^n}\right|^2 = \frac{1}{R2^n}\left(\frac{\sin(\pi Rrk/2^n)}{\sin(\pi rk/2^n)}\right)^2 . \tag{D4}$$

In the regime $r\ll 2^n$, it is convenient to introduce the continuum variable $x=\pi k/2^n\in[0,\pi]$ and approximate the distribution by the density

$$f(x|r) = \frac{1}{\pi R}\left(\frac{\sin(Rrx)}{\sin(rx)}\right)^2 . \tag{D5}$$

The QFT Fisher information with respect to the generator r is then

$$I_{\mathrm{QFT}}(r) = \int_0^\pi f(x|r) \left[\frac{\partial}{\partial r} \ln f(x|r)\right]^2 dx, \tag{D6}$$

with score function

$$\frac{\partial}{\partial r} \ln f(x|r) = 2Rx\cot(Rrx) - 2x\cot(rx). \tag{D7}$$

To obtain a more explicit form, we use the Fejer-kernel identity

$$\left(\frac{\sin(Rrx)}{\sin(rx)}\right)^2 = R + 2\sum_{m=1}^{R-1}(R-m)\cos(2mrx). \tag{D8}$$

Substituting this expansion into the integral above and using

$$\int_0^\pi x^2\, dx = \frac{\pi^3}{3} \tag{D9}$$

and

$$\int_0^\pi x^2 \cos(2jrx)\, dx = \frac{4\pi jr\cos(2\pi jr) + (4\pi^2 j^2 r^2 - 2)\sin(2\pi jr)}{(2jr)^3}, \tag{D10}$$

we obtain the exact series expression

$$I_{\mathrm{QFT}}(r) = \frac{8}{\pi R}\sum_{k=1}^{R-1}(R-k)\left[\frac{k\pi^3}{3} + \sum_{j=1}^{k-1}\frac{k-j}{j^3 r^3}\left(2\pi jr\cos(2\pi jr) + (2\pi^2 j^2 r^2 - 1)\sin(2\pi jr)\right)\right]. \tag{D11}$$

Keeping only the leading contribution in the large-r regime gives

$$I_{\mathrm{QFT}}(r) \approx \frac{4\pi^2}{9}(R^2 - 1). \tag{D12}$$

This expression provides the analytical QFT benchmark used in the main text.

E. Noise-robustness experiment details

We evaluate noise robustness using the previously trained 9- and 10-qubit HP-1 plus neural-decoder pipelines from the period-recovery study. For each system size, we use a measurement budget of $1024n^2$ samples per instance, giving 82944 measurements for $n = 9$ and 102400 for $n = 10$. The candidate periods are $r \in \{9, \dots, 80\}$ for 9 qubits and $r \in \{10, \dots, 99\}$ for 10 qubits. Probability distributions are generated from the exact subgroup support of the trained HP-1 circuit, and the neural decoder checkpoint is selected using noiseless validation accuracy. The selected checkpoints occur at epoch 3 with validation top-1 accuracy 0.9959 for 9 qubits, and at epoch 10 with validation top-1 accuracy 1.0000 for 10 qubits.

The evaluation set holds out 3 shifts for each candidate period and generates 8 independent noisy redraws for each held-out shift. This yields $72 \times 3 \times 8 = 1728$ test examples for 9 qubits and $90 \times 3 \times 8 = 2160$ test examples for 10 qubits. To probe robustness after the shallow quantum transformation, we insert a single global depolarizing channel after the full HP-1 circuit and before measurement. At the probability level, this is implemented as

$$\tilde{p}_\eta(x|r, s) = (1-\eta)p(x|r, s) + \eta\frac{1}{2^n}, \tag{E1}$$

where $p(x|r, s)$ is the ideal output distribution for period r and shift s, and η is the depolarizing strength. For each noisy distribution $\tilde{p}_\eta$, we resample $1024n^2$ outcomes and feed the resulting bit-string batch into the fixed neural decoder. The sweep uses the values

$$\eta \in \{1,\ 0.464,\ 0.215,\ 0.1,\ 0.0464,\ 0.0215,\ 0.01,\ 0.00464,\ 0.00215,\ 0.001\}. \tag{E2}$$

η	9-qubit accuracy	10-qubit accuracy
1.000	0.0139	0.0111
0.464	0.4913	0.8810
0.215	0.9948	1.0000
0.100	0.9954	1.0000
0.0464	0.9959	1.0000
0.0215	0.9942	1.0000
0.0100	0.9954	1.0000
0.00464	0.9959	1.0000
0.00215	0.9948	1.0000
0.00100	0.9948	1.0000

TABLE I. Held-out period-recovery accuracy under post-HP-1 global depolarizing noise.

Several features of Table I are worth noting. First, performance is essentially unchanged throughout the weak-to-moderate noise regime and remains near the noiseless baseline up to $\eta \approx 0.215$. Second, substantial degradation appears only once the noisy distribution is strongly mixed with the uniform distribution: at $\eta = 0.464$, the accuracy drops to 0.4913 for 9 qubits and 0.8810 for 10 qubits. Finally, when $\eta = 1$, the output distribution becomes exactly uniform and is therefore independent of both the period and the shift. In that limit, all subgroup information is erased, so the best possible behavior is chance-level guessing over the candidate periods. The observed accuracies 0.0139 and 0.0111 agree exactly with the corresponding chance levels 1/72 and 1/90, confirming that the numerical results are consistent with the depolarizing-noise model.